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A robust model predictive control unified framework for autonomous rendezvous and docking with a tumbling target

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Abstract

During autonomous rendezvous and docking (AR&D) with a tumbling target, multiple unfavorable engineering constraints exist, such as constraints on the control, velocity, or collision avoidance. Meanwhile, many sources of uncertainty exist, including incomplete and inaccurate measurements, control deviation, dynamic uncertainty, and inertial parameter uncertainty of the tumbling target, which significantly complicate the design of the guidance and control algorithm. Considering these challenges, this paper proposes a robust identification, planning, and control unified framework for AR&D with a tumbling target, which incorporates an extended Kalman filter (EKF) with a robust tube-based model predictive control for tracking (TMPT) controller. First, the EKF is designed to estimate the system state and inertial parameters of the target and then predict the future behavior of the tumbling target. Additionally, the prediction information of the target is transmitted to the designed TMPT controller as a reference signal. This TMPT controller unites trajectory planning and control in a single layer that can ensure the stability and recursive feasibility of the algorithm with a short prediction horizon. Moreover, it can guarantee that the closed-loop system is robustly and asymptotically convergent to the reachable set whose center is the optimal reachable periodic trajectory concerning the docking port of the tumbling target. The unified framework represents a robustness-to-optimality control strategy that can accomplish the AR&D mission efficiently while considering multiple engineering constraints and uncertainties. Finally, the efficiency of the proposed control framework is verified through a numerical simulation.
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Keywords: Autonomous rendezvous and docking; Unified framework; Extended Kalman filter; RMPC for tracking; Robustness-to-optimality control strategy

1. Introduction

Autonomous rendezvous and docking (AR&D) operations refer to the maneuvers required for a chaser/servicing spacecraft to arrive in the vicinity of a target spacecraft and establish a mechanical connection to it, which enables the chaser to accomplish on-orbit services such as repair, refueling, and upgrading of the target spacecraft (Lu & Liu, 2013; Flores-Abad et al., 2014; Ventura et al., 2016; Yu et al., 2016). Usually, the trajectory planning and control of AR&D missions consider strict engineering requirements regarding collision avoidance, velocity constraints, and control limitations (Luo et al., 2014; Ardaens et al., 2018). The spacecraft trajectory planning and control problem can be formulated as an optimal control problem (Chai et al., 2019). Generally, the optimal control problem or trajectory optimization problem of the AR&D mission is solved using numerical methods that are typically classified

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into two categories: indirect and direct methods. Methods of the former category derive the first-order necessary conditions for optimal control problems using the Pontryagin maximum principle. Therefore, they have advantages such as high accuracy and compatibility with online computation (Boyarko et al., 2011; Yang & Baoyin, 2015). However, deriving the first-order necessary conditions is nontrivial and unsuitable for cases with complicated constraints, such as constraints on collision avoidance and control saturation. On the other hand, direct methods discretize and parametrize the control input and state to transform the optimal problem into a nonlinear programming problem (NLP). Direct methods are straightforward and accessible; they can also handle constraints explicitly, and as such, they have attracted increasing attention from engineers. However, the classic direct method is unsuitable for online and autonomous implementations if the trajectory optimization problem is highly constrained (Lu & Liu, 2013). Therefore, many researchers have attempted to improve the efficiency of the classic method for trajectory planning in AR&D missions. Some researchers have proposed guidance methods that combine an artificial potential function with fuzzy logic in AR&D missions (Zhang et al., 2010; Badawy & McInnes, 2008). However, this method may not find a global minimum. Lu et al. (2013) transformed the NLP into a second-order cone programming problem (SOCP) using a lossless relaxation technique to enable autonomous onboard implementation. However, the SOCP solution may not converge to a feasible trajectory in a general nonlinear case (Leomanni et al., 2022). Ventura et al. (2016) proposed a fast and near-optimal guidance method for AR&D missions based on a dynamic inversion strategy. However, the methods mentioned above attempt to drive the state of the chaser to a specific terminal point/set, thus requiring more decision variables and may not be suitable for online implementation, even for short-range AR&D missions. Charles et al. (2021) developed a reinforcement learning-based control method for AR&D with a rotating target. However, it is unsuitable for cases with complicated state and control input constraints.

Compared with the previous methods, model predictive control (MPC) is a finite-horizon and receding optimization method that can reduce the number of decision variables. Moreover, the ability to handle constraints systematically and elegantly and use a wide range of prediction models makes MPC popular in space engineering, such as for AR&D missions. In earlier studies, some researchers (Richards & How, 2003; Richards & How, 2006; Hartley et al., 2012; Weiss et al., 2015) investigated the MPC strategy for controlling spacecraft in various rendezvous scenarios with a static docking port, considering multiple constraints such as the thrust magnitude, line-of-sight (Li et al., 2017), collision avoidance(Zagaris et al., 2018; Wang et al., 2023) and even the chance constraints (Gavilan et al., 2012). An explicit MPC approach has also been applied to the AR&D mission to improve the online implementation of the controller (Leomanni et al., 2014). In recent years, the problem of AR&D with a non-cooperative target has attracted increasing attention. Compared with a cooperative target, a non-cooperative target has some uncontrolled attitude dynamics behaviors (e.g., rotating or tumbling) and/or uncertainties (e.g., unknown mass and moment of inertia). Some researchers have applied the MPC method to approach a rotating platform (Di Cairano et al., 2012; Park et al., 2017). Li et al. (2017) extended the application of MPC to a 3-D space rendezvous case. Corpino et al. (2018) designed an MPC controller to approach a tumbling target while estimating the rotational dynamic state of the target. To handle the uncertainties in the AR&D mission process, some researchers (Carson, 2008; Zhu et al., 2018) have applied a robust MPC strategy to trajectory planning and control of the AR&D mission. To further improve the algorithm's robustness, some researchers have designed tube-based robust MPC strategies to approach a tumbling target with uncertainties, considering different scenarios and engineering demands (Mammarella et al., 2018; Buckner et al., 2018; Dong et al., 2020; Oestreich et al., 2023; Specht et al., 2023).

However, the MPC controllers proposed for AR&D missions in previous studies could be further improved. First, suppose that the target's initial position is far from that of the chaser. In this case, the proposed MPC controllers are neither stable nor robust; they are not recursively feasible or perform global optimization to guarantee feasibility. Second, if the reference signal changes, the proposed MPC controllers will likely lose their feasibility (they cannot find a feasible solution) and stability, which is the case with the time-varying position of the docking point studied herein. Third, the future information of the tumbling target, such as its docking point position, was assumed to be known a priori in previous studies. However, the future behavior of a tumbling target cannot be accurately predicted, considering the multiple uncertainties (i.e., inertial parameters) and inaccurate/incomplete measurements in actual engineering practice.

The U.S. National Research Council considers a reliable and robust guidance algorithm for AR&D with uncontrolled tumbling objects as a high-priority technology for the 2012–2021 decade and beyond. This demands that the next-generation guidance algorithm have the following fundamental properties: optimality, robustness, and real-time implementation (Ventura et al., 2016). To meet these requirements and overcome the drawbacks of the MPC controllers proposed in previous studies, this paper proposes a robust identification, prediction, motion planning, and control unified framework for AR&D with a tumbling target under engineering constraints and multiple uncertainties. This approach can accomplish the AR&D mission efficiently, autonomously, and safely. The main contributions of this study are summarized as follows:

1) The proposed unified framework incorporates an extended Kalman filter (EKF) with a robust tube-based model predictive control for tracking (TMPT) controller, which can reduce uncertainties during the control process and then perform the AR&D mission successfully with high efficiency. In addition, the framework represents a robustness-to-optimality control strategy that can improve tracking accuracy and reduce uncertainties compared to traditional methods.
2) The proposed TMPT controller is designed based on MPC for tracking theory (Limon et al., 2008; Limon et al., 2016), and a feedback control strategy that uses the trajectory of the tumbling target as the reference to be followed, uniting dynamic trajectory planning and control in a single layer to improve the real-time performance of the algorithm. Moreover, through a comprehensive design of the artificial steady state, control input, and terminal constraint, the controller guarantees that the closed-loop system is stable and recursive feasible and converges robustly and asymptotically to the optimal admissible periodic trajectory concerning the tumbling target' docking port.
3) The stability and feasibility of the proposed TMPT controller can be ensured with a small prediction horizon up to the system order, even if the chaser is far from the tumbling target. Thus, it can reduce the computational complexity of the control algorithm, making it suitable for online implementation. Moreover, the constraints in the proposed TMPT controller are independent of the reference signal, allowing for sudden changes in the reference signals without compromising feasibility or stability.

The remainder of this paper is organized as follows. Section 2 presents the problem formulation, including the dynamic model, engineering constraints, control object and formulation of the optimal control, and uncertainty analysis. Section 3 details the unified control framework design for the AR&D mission, including the design of the EKF and the TMPT controller. The simulation results and corresponding discussion are presented in Section 4. Finally, Section 5 presents the main conclusions.

2. Docking dynamics and problem formulation

In this study, we developed a robust and near-optimal trajectory planning and control algorithm to achieve a 3-DOF docking mission with a tumbling target with unknown inertial parameters (behaving uncontrolled). First, the dynamic models of translational and rotational motion between the chaser and target in the close-range AR&D mission are summarized. Then, the optimal and robust trajectory planning and control problem is formulated, considering practical engineering constraints and uncertainties. Fig. 1 shows the geometry and coordinate

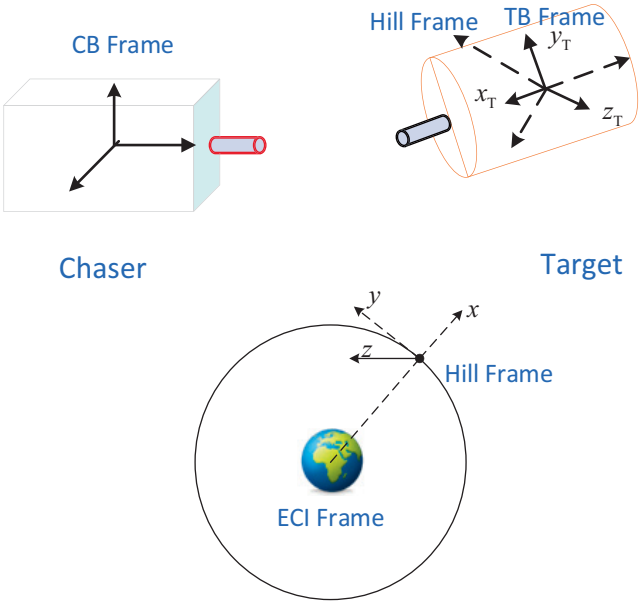


Fig. 1. Spacecraft geometry and coordinate frames.

frames of the chaser and target. The coordinate frames are defined as follows:

1) Earth-centered inertial coordinate frame (ECI): The origin of the ECI frame coincides with the center of the Earth; the x-axis points to the vernal equinox, the z-axis is along the North Pole, and the y-axis completes the right-hand rule.
2) Hill's frame (local vertical local horizontal frame, LVLH): The origin of the ECI frame coincides with the center of mass of the target; the x-axis aligns with the radial direction, the y-axis is along the velocity direction of the target, and the z-axis completes the right-hand rule.
3) Body frame: The origin of the body frame is the center of mass of the spacecraft. The axes x_C,y_C and y_C of the chaser's body frame are aligned with their principal body axes, denoted as the CB frame. The x_T-axis

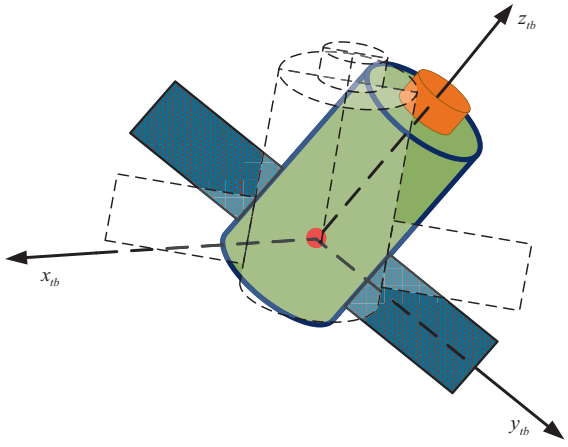


Fig. 2. Tumbling target with inertial parameters uncertainty.

of the target points towards the head of the target, the y_T-axis points to the right side of the chaser (when looking in the direction of the x_T-axis) perpendicular to the x_T-axis, and the z_T-axis completes the right-hand rule, denoted as TB frame. Fig. 2.

2.1. Dynamic model of the AR&D maneuver

For close-range AR&D missions, the linearized Clohessy-Wiltshire (C-W) equations are usually utilized to describe the relative translational motion between two spacecraft as follows (Shao et al., 2023):

$$\begin{cases} \ddot{x} = 2n\dot{y} + 3n^2x + u_x + d_x \\ \ddot{y} = -2n\dot{x} + u_y + d_y \\ \ddot{z} = -n^2z + u_z + d_z \end{cases} \tag{1}$$

where the vectors $\boldsymbol{r} = [xyz]^{\mathrm{T}}$ and $\dot{\boldsymbol{r}} = [\dot{x}\dot{y}\dot{z}]^{\mathrm{T}}$ are the relative position and velocity of C.M.(center of mass) of the two spacecraft expressed in the Hill frame, respectively. The vectors,$\boldsymbol{u} = [u_xu_yu_z]^{\mathrm{T}}$ and $\boldsymbol{d} = [d_xd_yd_z]^{\mathrm{T}}$ are the control input and external disturbance, respectively. $\boldsymbol{\Omega} = [00n]^{\mathrm{T}}$ is the orbital angular velocity of the LVLH frame concerning the ECI frame, expressed in the LVLH frame, where $n = \sqrt{\mu/r_t^3}$, μ and r_t are the gravitational constant and radius of the target's orbit, respectively. To facilitate controller design, Eq. (1) can be rewritten as

$$\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{u} + \boldsymbol{B}\boldsymbol{d} \tag{2}$$

where the state $\boldsymbol{x}$ is defined as $\boldsymbol{x} = [\boldsymbol{r}^{\mathrm{T}}\dot{\boldsymbol{r}}^{\mathrm{T}}]^{\mathrm{T}} \in \mathbb{R}^6$. The matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ are defined as

$$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{0}_3 & \boldsymbol{I}_3 \\ \boldsymbol{K}(n) & -2\boldsymbol{\Omega} \end{bmatrix}, \boldsymbol{B} = \begin{bmatrix} \boldsymbol{0}_3 \\ \boldsymbol{E}_3 \end{bmatrix} \tag{3}$$

where $\boldsymbol{E}_3$ and $\boldsymbol{0}_3$ are the identity matrix and zero matrix of size 3, respectively.$\boldsymbol{S}(\boldsymbol{\Omega})$ is a cross-product matrix of the orbital angular velocity $\boldsymbol{\Omega}$, and $\boldsymbol{K}(n)$ is defined as follows,

$$\boldsymbol{K}(n) = \begin{bmatrix} 3n^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -n^2 \end{bmatrix} \tag{4}$$

The continuous system(Eq.(2)) is discrete over the sampling time T_s using the zero-order hold (ZOH) method,

$$\boldsymbol{x}(k+1) = \boldsymbol{A}_d\boldsymbol{x}(k) + \boldsymbol{B}_d\boldsymbol{u}(k) + \boldsymbol{\omega} \tag{5}$$

where $\boldsymbol{A}_d$ and $\boldsymbol{B}_d$ are the corresponding state and input matrices of the discrete-time system. $\boldsymbol{\omega} = \boldsymbol{B}_d \cdot \boldsymbol{d} \in \mathbb{R}^6$ is the uncertainty caused by the external disturbance or uncertainties.

Define a quaternion as the form $\boldsymbol{q} = [\boldsymbol{q}_v^{\mathrm{T}}q_0]^{\mathrm{T}}$, where q_0 is a scalar and $\boldsymbol{q}_v \in \mathbb{R}^3$ is a vector. The quaternion multiplication operator $[\boldsymbol{q}\otimes]$ is defined as (Ge et al., 2020),

$$[\boldsymbol{q}\otimes] = \begin{bmatrix} q_0\boldsymbol{I}_3 - S(\boldsymbol{q}_v) & \boldsymbol{q}_v \\ -\boldsymbol{q}_v^{\mathrm{T}} & q_0 \end{bmatrix} \tag{6}$$

where $S(\boldsymbol{q}_v)$ is a cross-product matrix of $\boldsymbol{q}_v$.

The kinematic equation of the tumbling target is expressed in quaternion form and can be written as

$$\dot{\boldsymbol{q}} = \frac{1}{2}\,\bar{\boldsymbol{\omega}}\otimes\boldsymbol{q} \tag{7}$$

where $\bar{\boldsymbol{\omega}} = [\boldsymbol{\omega}^{\mathrm{T}}0]^{\mathrm{T}}$ and $\boldsymbol{\omega} = [\omega_x\omega_y\omega_z]^{\mathrm{T}}$ denote the target's angular velocity relative to the inertial ECI frame, defined in the TB frame. The quaternion $\boldsymbol{q}$ represents the attitude quaternion rotating from the inertial ECI to the TB frame. We define I_x, I_y, and I_z as the principal inertial moments of the target. Generally, the Euler equation is used to describe the rotational dynamics of a tumbling target, as follows:

$$\begin{cases} I_x\dot{\omega}_x - (I_y - I_z)\omega_y\omega_z = 0 \\ I_y\dot{\omega}_y - (I_z - I_x)\omega_z\omega_x = 0 \\ I_z\dot{\omega}_z - (I_x - I_y)\omega_x\omega_y = 0 \end{cases} \tag{8}$$

2.2. Engineering constraints

AR&D missions include certain practical engineering constraints, such as constraints on control saturation, approaching velocity, and collision avoidance.

2.2.1. Control saturation

The control input vector $\boldsymbol{u} = [u_xu_yu_z]^{\mathrm{T}}$ is considered as the control signal. Due to the finite thrust capability of the chaser, a maximum value $u_{\max}$ exists for each control component. Then, the control saturation is defined as follows:

$$\| \boldsymbol{u} \|_\infty \le u_{\max} \tag{9}$$

To facilitate the controller design, Eq.(9) is rewritten in the following form:

$$\begin{bmatrix} \boldsymbol{E}_3 \\ -\boldsymbol{E}_3 \end{bmatrix} \boldsymbol{u} \le \boldsymbol{u}_{\max} \tag{10}$$

where, $\boldsymbol{u}_{\max} = u_{\max} \cdot \boldsymbol{I}_{6\times1}$, $\boldsymbol{I}_{6\times1}$ denotes a 6-dimensional column vector of unit elements.

2.2.2. Approaching velocity constraint

To ensure the safety of docking operations and avoid possible collisions in case an emergency occurs, the approaching velocity constraint is set as

$$\| \dot{\boldsymbol{r}} \|_\infty \le v_{\max} \tag{11}$$

where $v_{\max}$ denotes the maximum approaching velocity value for each direction. Similarly, Eq.(11) can be rewritten in linearized form as follows:

$$\begin{bmatrix} \boldsymbol{E}_3 \\ -\boldsymbol{E}_3 \end{bmatrix} \dot{\boldsymbol{r}} \le \boldsymbol{v}_{\max} \tag{12}$$

In summary, the control saturation and velocity constraints are the time-wise constraints on the state and control input and can be rewritten in the compact form:

$$(\boldsymbol{x}(k), \boldsymbol{u}(k)) \in \mathbb{Z} \tag{13}$$

where $\mathbb{Z}$ is a suitable bounded set obtained from Eqs.(10) and(12). Furthermore, additional constraints posed as a linear combination of state and control input can easily be added to this framework.

2.2.3. Collision avoidance constraint

During the close-range AR&D mission, a "keep-out" zone is considered critical to avoid a possible collision with the target. Generally, a target-center sphere with a radius r_s that can enclose the target is chosen to describe the "keep-out" zone. Then, the "keep-out" zone is defined as follows:

$$d = -\| \boldsymbol{r} \|_2^2 + r_s^2 \leq 0 \tag{14}$$

To reduce the computational burden of the onboard implementation, Eq.(14) is linearized at the position of the docking port $\boldsymbol{r}_d = [x_d y_d z_d]^{\mathrm{T}}$:

$$- [x_d y_d z_d]\boldsymbol{r} \leq -r_s^2 \tag{15}$$

Then, the collision avoidance constraint is then expressed as a polytopic representation as follows:

$$\Gamma_{kz}^{(k)} = \left\{ \boldsymbol{r} \in \mathbb{R}^3 | - [x_d y_d z_d]\boldsymbol{r} \leq -r_s^2 \right\} \tag{16}$$

Eq.(16) is a time-varying and hard constraint for safety considerations, bringing significant challenges to the controller design.

2.3. Control objective and optimal control problem formulation

The objective of the controller is to drive the chaser to approach (the neighborhood of) the docking port of the target and to maintain synchronization with the target, concerning both position and attitude at the end of the maneuvers. Moreover, the deviation of the linear and angular velocities between the chaser and target should also be eliminated or within a small range to achieve smooth operation of the AR&D mission. The terminal docking-enabling condition can be expressed as follows:

$$\begin{cases} \boldsymbol{\rho}_p \approx \boldsymbol{R}_{OT}\boldsymbol{r}_d \\ \boldsymbol{\rho}_v \approx (\boldsymbol{R}_{OT}\boldsymbol{\omega}_T - \bar{\boldsymbol{\omega}}) \times \boldsymbol{R}_{OT}\boldsymbol{r}_d \\ \boldsymbol{R}_{IC} \approx \boldsymbol{R}_{IT} \\ \boldsymbol{R}_{IC}\omega_C \approx \boldsymbol{R}_{IT}\boldsymbol{\omega}_T \end{cases} \tag{17}$$

where $\boldsymbol{\rho}_p$ and $\boldsymbol{\rho}_v$ are the expected terminal relative position and velocity of the chaser, respectively. $\bar{\boldsymbol{\omega}}$ is the orbital angular velocity of the tumbling target in the ECI frame. $\boldsymbol{R}_{OT}$ is the direction cosine matrix that can transform a vector from the BF frame into the Hill frame. $\boldsymbol{R}_{IC}$ and $\boldsymbol{R}_{IT}$ are directional cosine matrices that transform a vector from the CB and TB frames into the ECI frame.

In engineering practice, the orientation control problem can be effectively decoupled from translation owing to the reaction control system (Caubet et al., 2015). Furthermore, the amount of fuel used to drive the chaser to the docking port of the target is far greater than that used for attitude synchronization during the chaser's final approach (Sternberg, 2017). Therefore, for simplicity, this study did not consider the problem of orientation synchronization (Caubet et al., 2015; Li et al., 2017; Dong et al., 2020). This study investigated the AR&D problem under the aforementioned engineering constraints and multiple uncertainties, requiring a guidance and control algorithm with the essential features of optimality, robustness, real-time performance, and high efficiency. The specific requirements for the guidance are as follows:

1) Optimality: The guidance and control algorithm can find the optimal or near-optimal (based on the performance index, e.g., energy consumption is minimized and tracking accuracy is maximized) and reachable nominal trajectory and corresponding control inputs that can steer the chaser from the initial state to achieve the docking-enabling conditions.
2) Robustness: The guidance and control algorithms should ensure that the state and control inputs of the system satisfy engineering constraints under uncertainties to guarantee the safety and success of the AR&D mission.
3) Real-time execution: The guidance and control algorithms should have low computational complexity for real-time and autonomous online implementation. Moreover, when the target is far away or sudden changes occur, the algorithm should still ensure stability and feasibility and replan the trajectory in an autonomous and real-time manner.
4) High efficiency: The guidance and control algorithm should identify and reduce uncertainties (i.e., the inertial parameters of the tumbling target) during the AR&D mission, thus improving the algorithm's performance (i.e., saving rendezvous time and fuel consumption).

2.4. Uncertainty analysis

The AR&D mission process has multiple uncertainties, including dynamic uncertainty, control deviation, and estimation error. The dynamic uncertainty usually includes the J_2 gravitational force, gravity gradient, atmospheric drag, linearization error, and unknown or unmodeled disturbances, which are bounded and denoted as $\boldsymbol{d} = [d_x d_y d_z]^{\mathrm{T}}$. The dynamic uncertainty of the full state is bounded by a polytopic set $\mathbb{W}_1$ as follows:

$$\mathbb{W}_1 = \left\{ \boldsymbol{\omega} = \boldsymbol{B}_{w_1} \cdot \boldsymbol{d} \in \mathbb{R}^6 | \| \boldsymbol{d} \|_\infty \leq d_\omega \right\} \tag{18}$$

According to the literature (Fehse, 2003), the linearization error of the C-W equations can be neglected if close-range operations are considered.

In engineering practice, a control deviation $\boldsymbol{d}_u$ may exist between the actual control input $\boldsymbol{u}(k)$ and the nominal con-

trol input $\bar{\boldsymbol{u}}(k)$. For simplicity, the control deviation $\boldsymbol{d}_u$ is assumed to be linear with $\bar{\boldsymbol{u}}(k)$, as follows:

$$\boldsymbol{u}(k) = \bar{\boldsymbol{u}}(k) + \boldsymbol{d}_u, \boldsymbol{d}_u = -\alpha(k)\,\bar{\boldsymbol{u}}(k) \tag{19}$$

where, $\alpha(k) \in (\underline{\alpha}, \bar{\alpha})$.

Thus, the effect of the control deviation on the system is bounded by a polytopic set $\mathbb{W}_2$, as follows:

$$\mathbb{W}_2 = \left\{\boldsymbol{\omega} = \boldsymbol{B}_d \cdot \boldsymbol{d}_u \in \mathbb{R}^6 | \| \boldsymbol{d}_u \|_\infty \le u_{\max} \cdot \alpha(k), \alpha(k) \in (\underline{\alpha}, \bar{\alpha})\right\} \tag{20}$$

In addition, the measurement/estimated error of the full state is also bounded (or a certain probability) by a polytopic set $\mathbb{W}_3$, as follows:

$$\mathbb{W}_3 = \left\{\boldsymbol{\omega} = \beta(k) \cdot \hat{\boldsymbol{x}} \in \mathbb{R}^6 | \beta(k) \in (\underline{\beta}, \bar{\beta})\right\} \tag{21}$$

where $\hat{\boldsymbol{x}}$ is the estimated full state.

Then, the sum of the uncertainty sets is denoted as

$$\mathbb{W} = \mathbb{W}_1 \oplus \mathbb{W}_2 \oplus \mathbb{W}_3 \tag{22}$$

where, $\oplus$ represents the Minkowski sum. Because $\mathbb{W}_1$, $\mathbb{W}_2$ and $\mathbb{W}_3$ are compact and convex polytopes, the uncertainty set $\mathbb{W}$ is a bounded polytopic set and is expressed as

$$\mathbb{W} = \left\{\boldsymbol{\omega} \in \mathbb{R}^6 | \boldsymbol{A}_w \cdot \boldsymbol{\omega} \le \boldsymbol{b}_w\right\} \tag{23}$$

2.5. Nominal model

For a system modeled using Eq.(5), the corresponding nominal system is defined as

$$\bar{\boldsymbol{x}}(k+1) = \boldsymbol{A_d}\,\bar{\boldsymbol{x}}(k) + \boldsymbol{B}_d\,\bar{\boldsymbol{u}}(k) \tag{24}$$

Given the initial state $\bar{\boldsymbol{x}}(k)$ of the nominal system and the corresponding control input sequence $\bar{\boldsymbol{U}}(k) = [\bar{\boldsymbol{u}}(k|k), \bar{\boldsymbol{u}}(k+1|k), \ldots, \bar{\boldsymbol{u}}(k+N_p-1|k)]$, the N_p steps ahead of the predicted state's trajectory are obtained as $\bar{\boldsymbol{X}} = [\bar{\boldsymbol{x}}(k|k), \bar{\boldsymbol{x}}(k+1|k), \ldots, \bar{\boldsymbol{x}}(k+N_p|k)]$.

2.6. Constraint tightening for the nominal system

Because uncertainties exist in real environments, the evolution of the real system's trajectory will not coincide with the nominal one. Choosing a control input that includes the nominal and the feedback inputs is sensible to reduce the effects of uncertainties. The nominal one is capable of driving the nominal state to the terminal point/set while the feedback one can force the actual state's trajectory to lie close to the nominal state's trajectory, as described in the literature (Limon et al., 2010):

$$\boldsymbol{u}(k) = \bar{\boldsymbol{u}}(k|k) + \boldsymbol{K}(\boldsymbol{x}(k) - \bar{\boldsymbol{x}}(k|k)) \tag{25}$$

where $\boldsymbol{u}(k) \in \mathbb{U}$ is the nominal control input, $\boldsymbol{K}(\boldsymbol{x}(k) - \bar{\boldsymbol{x}}(k|k))$ is the feedback control input and $\mathbb{U}$ is a polytope.

From Eqs.(5), (24), and (25), the error dynamics can be derived:

$$\boldsymbol{e}(k+1) = \boldsymbol{A}_k \boldsymbol{e}(k) + \boldsymbol{\omega}, \text{ with } \boldsymbol{A}_k = \boldsymbol{A}_d + \boldsymbol{B}_d \boldsymbol{K} \tag{26}$$

where $\boldsymbol{e}(k) = \boldsymbol{x}(k) - \bar{\boldsymbol{x}}(k) \in \mathbb{W}$ is the error between the actual state and the nominal state.

If the matrix $\boldsymbol{A}_k$ is Hurwitz, the evolution of $\boldsymbol{e}(k)$ is bounded:

$$\boldsymbol{A}_k \mathbb{W} \oplus \boldsymbol{\omega} \subseteq \mathbb{W}, \ \boldsymbol{\omega} \in \mathbb{W} \tag{27}$$

Thus, the actual trajectory $\boldsymbol{X}$ lies in the neighborhood of the nominal trajectory $\bar{\boldsymbol{X}}(k)$, which is called the reachable set, and is given by

$$\mathbb{H}(i) = \bigoplus_{j=0}^{i} \boldsymbol{A}_k^j \mathbb{W} \tag{28}$$

The reachable set $\mathbb{H}(i)$ quantifies the propagation of uncertainty. Considering that $\boldsymbol{e}(k) = \boldsymbol{x}(k) - \bar{\boldsymbol{x}}(k) \in \mathbb{W}$ for any possible realization of uncertainties, the trajectory of the uncertainty system under the control of the control law (Eq.(25)) is such that

$$\boldsymbol{x}(k+i) = \bar{\boldsymbol{x}}(k+i|k) + \boldsymbol{s}(k+i|k), i = 0, \ldots, N_p \tag{29}$$

where, $\boldsymbol{s}(k+i|k) \in \mathbb{H}(i)$. The set of reachable sets $\mathbb{H}(i)$ is called a tube. In addition, if the initial condition $\boldsymbol{x}(k) = \bar{\boldsymbol{x}}(k) \oplus \mathbb{W}$ holds and the nominal control sequence is $\bar{\boldsymbol{U}}(k)$, such that:

$$\bar{\boldsymbol{x}}(k+i|k) \in \mathbb{X} \ominus \mathbb{H}(i), i = 0, \ldots, N_p \tag{30}$$

where $\ominus$ denotes the Pontryagin set difference and $\mathbb{X}$ is the state's constraint, then for any possible realization of the disturbance, the trajectory and controls derived from the uncertain system (Eq.(5)) and control law (Eq.(25)) satisfy,

$$\boldsymbol{x}(k+i|k) \in \mathbb{X}, i = 0, \ldots, N_p \tag{31}$$

Generally, it is not easy to accurately calculate the values of the reachable sets $\mathbb{H}(i)$. However, when the constraint sets of the state and control are polytopes, some approaches can be used to evaluate it efficiently, including the supporting function method (Rakovic et al., 2005) and the zonotope method (Alvarado et al., 2007). Next, the supporting function method is described in detail.

Definition 1. (Kolmanovsky Gilbert, 1998): The support function of a set $\mathbb{S} \subset \mathbb{R}^n$ evaluated at $\boldsymbol{\eta} \in \mathbb{R}^n$ is $h_{\mathbb{S}}(\boldsymbol{\eta}) = \sup_{s \in \mathbb{S}} \boldsymbol{\eta}^T \boldsymbol{s}$.

Suppose $\mathbb{X}$ is a polytope, where $\mathbb{X} = \{\boldsymbol{x} \in \mathbb{R}^n | \boldsymbol{G} \cdot \boldsymbol{x} \le \boldsymbol{h}, \boldsymbol{G} \in \mathbb{R}^{m \times n}, \boldsymbol{h} \in \mathbb{R}^{m \times 1}\}$. Then, the following expressions are equivalent:

$$\begin{aligned} \boldsymbol{G} \cdot \boldsymbol{x}(k) \le \boldsymbol{h} \Longleftrightarrow \boldsymbol{G} \cdot (\bar{\boldsymbol{x}}(k) + \boldsymbol{s}) \le \boldsymbol{h}, \forall \boldsymbol{s} \\ \in \mathbb{H}(i) \Longleftrightarrow \boldsymbol{G} \cdot \bar{\boldsymbol{x}}(k) \le \boldsymbol{h} - \max_{\boldsymbol{s} \in \mathbb{H}(i)} \boldsymbol{G} \cdot \boldsymbol{s} \end{aligned} \tag{32}$$

Now, the key point is determining how to calculate $\max_{s\in\mathbb{H}(i)} \boldsymbol{G}\cdot\boldsymbol{s}$, where $\boldsymbol{G}$ is row-wise, and $\boldsymbol{G}_i^T \subset \mathbb{R}^{1\times n}$ is the i - th row of the matrix $\boldsymbol{G}$.

$$\max_{s\in\mathbb{H}(i)} \boldsymbol{G}_i\cdot\boldsymbol{s} = h_{\mathbb{H}(i)}(\boldsymbol{G}_i) = h_{\bigoplus_{j=0}^{i} \boldsymbol{A}_k^j\mathbb{W}}(\boldsymbol{G}_i) = \sum_{j=0}^{i} h_{\mathbb{W}}((\boldsymbol{A}_k^j)^{\mathrm{T}}\cdot\boldsymbol{G}_i) \tag{33}$$

where $h_{\mathbb{W}}((\boldsymbol{A}_k^j)^{\mathrm{T}}\cdot\boldsymbol{G}_i)$ can be easily computed using linear programming. We can then obtain the nominal constraints using Eq.(32).

Thus, the nominal constraints on the control saturation and approaching velocity obtained using the tightening constraint of Eq .(32) are defined as

$$\bar{\boldsymbol{x}}(k+i|k)\in\bar{\mathbb{X}}\triangleq\mathbb{X}\ominus\mathbb{H}(i),\bar{\boldsymbol{u}}(k+i|k) \in \bar{\mathbb{U}}\triangleq\mathbb{U}\ominus\boldsymbol{K}\mathbb{H}(i), i=0,\ldots,N_p \tag{34}$$

where, $\bar{\mathbb{X}}$ and $\bar{\mathbb{U}}$ has the polytopic representation.

Similarly, the nominal collision avoidance constraint for the nominal system obtained using the tightening constraint of Eq.(16) is denoted as

$$\bar{\Gamma}_{kz}^{(k)} = \left\{\bar{\boldsymbol{\rho}}\in\mathbb{R}^3 \mid -(x_d,y_d,z_d)\,\bar{\boldsymbol{\rho}}\le -\bar{r}_s^2\right\} \tag{35}$$

3. Controller design

In this study, a robust identification, planning, and control unified framework was formulated for AR&D with a tumbling target, considering some practical engineering constraints, the target's inertial parameters, and the additive uncertainties of the system dynamics, as shown in Fig. 3.

First, the states required by the controller design are estimated using the extended Kalman filter (EKF), including the inertial parameters of the tumbling target. Additionally, the estimated inertial parameters are utilized to predict the future behavior of the tumbling target, such as the relative position of the docking port. Finally, the controller is designed based on MPC for tracking theory, which guarantees that the nominal state of the closed system converges asymptotically to the optimal reachable trajectory. The tube-based feedback control strategy forces every realization of the actual states to remain in a reachable set (the set of reachable sets is called a tube), whose center is the nominal trajectory. Moreover, parameter estimation/identification occurs at each sampling time; therefore, the uncertainties decrease during the control process of the AR&D mission, thus reducing conservatism and enhancing the performance of the proposed algorithm.

3.1. Extended Kalman filter

This study estimates the attitude of the tumbling target expressed by a quaternion $\hat{\boldsymbol{q}}$, angle velocity $\hat{\boldsymbol{\omega}}$, and inertial parameters $\hat{\boldsymbol{p}}$, as well as the relative position $\hat{\boldsymbol{r}}$ and relative velocity $\hat{\boldsymbol{v}}$ using the EKF, assuming that the information of attitude quaternion $\boldsymbol{q}$ and relative position $\boldsymbol{r}$ can be measured by sensors. Therefore, the attitude of the tumbling target can be predicted based on its estimated state.

Then, the inertial parameters $\boldsymbol{p}=[p_x p_y p_z]^{\mathrm{T}}$ are defined as follows

$$p_x=\frac{I_y-I_z}{I_x}, p_y=\frac{I_z-I_x}{I_y}, p_z=\frac{I_x-I_y}{I_z} \tag{36}$$

Substituting Eq. (36) into Eq. (7), the rotational dynamics can be formulated as,

$$\dot{\boldsymbol{\omega}}=\boldsymbol{\psi}(\boldsymbol{\omega},\boldsymbol{p})=\begin{bmatrix} p_x\omega_y\omega_z \\ p_y\omega_x\omega_z \\ p_z\omega_x\omega_y \end{bmatrix} \tag{37}$$

Linearizing Eq.(37) at the estimated angle velocity $\hat{\boldsymbol{\omega}}$ and the inertial ratio $\hat{\boldsymbol{p}}$, as follows

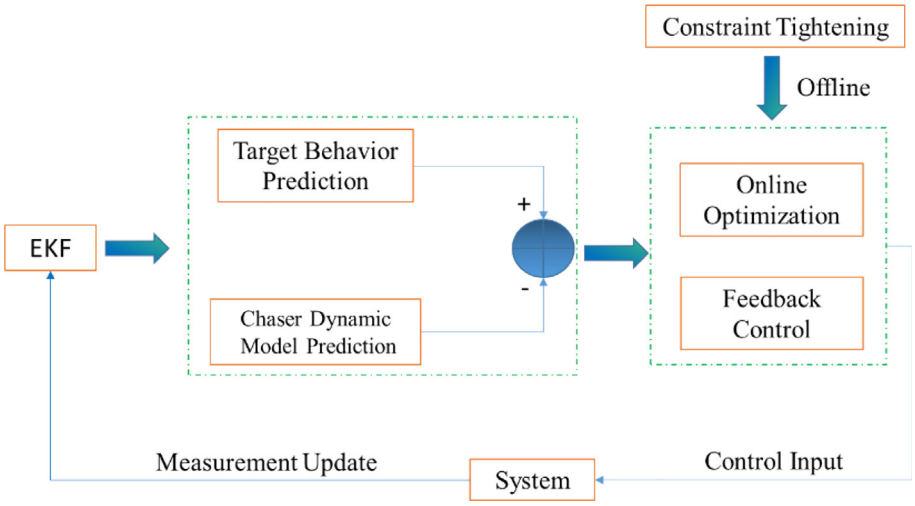


Fig. 3. Block diagram of proposed control method.

$$\frac{d}{dt}\delta\boldsymbol{\omega} = \boldsymbol{M}(\hat{\boldsymbol{\omega}},\hat{\boldsymbol{p}})\delta\boldsymbol{\omega} + \boldsymbol{N}(\hat{\boldsymbol{\omega}},\hat{\boldsymbol{p}})\delta\boldsymbol{p} \tag{38}$$

where

$$\begin{aligned}\boldsymbol{M}(\hat{\boldsymbol{\omega}},\hat{\boldsymbol{p}}) &= \frac{\partial\boldsymbol{\psi}}{\partial\boldsymbol{\omega}} = \begin{bmatrix} 0 & \hat{p}_x\hat{\omega}_z & \hat{p}_x\hat{\omega}_y \\ \hat{p}_y\hat{\omega}_z & 0 & \hat{p}_y\hat{\omega}_x \\ \hat{p}_z\hat{\omega}_y & \hat{p}_z\hat{\omega}_x & 0 \end{bmatrix}, \\ \boldsymbol{N}(\hat{\boldsymbol{\omega}},\hat{\boldsymbol{p}}) &= \frac{\partial\boldsymbol{\psi}}{\partial\boldsymbol{p}} = \begin{bmatrix} \hat{\omega}_y\hat{\omega}_z & 0 & 0 \\ 0 & \hat{\omega}_x\hat{\omega}_z & 0 \\ 0 & 0 & \hat{\omega}_x\hat{\omega}_y \end{bmatrix}\end{aligned} \tag{39}$$

We define the state vector of the EKF as $\boldsymbol{x}_e = [\boldsymbol{q},\boldsymbol{\omega},\boldsymbol{p},\boldsymbol{r},\boldsymbol{v}]^T$. Then, the following conditions hold:

$$\begin{cases} \dot{\boldsymbol{q}} = -\frac{1}{2}\boldsymbol{q}\otimes\bar{\boldsymbol{\omega}} \\ \dot{\boldsymbol{\omega}} = d(\boldsymbol{\omega},\boldsymbol{p}) + B(\boldsymbol{p})\varepsilon_\tau \\ \dot{\boldsymbol{p}} = 0 \\ \dot{\boldsymbol{r}} = 0 \\ \dot{\boldsymbol{v}} = \ddot{\boldsymbol{r}} = -2\boldsymbol{S}(\boldsymbol{\Omega})\dot{\boldsymbol{r}} + \boldsymbol{K}(n)\boldsymbol{r} \end{cases} \tag{40}$$

where the rotational control part of the system matrix is given by

$$\boldsymbol{F}_1 = \begin{bmatrix} \boldsymbol{0}_4 & \frac{1}{2}\boldsymbol{\Gamma} & \boldsymbol{0}_{4\times3} \\ \boldsymbol{0}_{3\times4} & \boldsymbol{M} & \boldsymbol{N} \\ \boldsymbol{0}_{3\times4} & \boldsymbol{0}_3 & \boldsymbol{0}_3 \end{bmatrix} \tag{41}$$

where,

$$\boldsymbol{\Gamma} = \begin{bmatrix} -\boldsymbol{q}_v^T \\ q_0\boldsymbol{I}_3 + S(\boldsymbol{q}_v) \end{bmatrix} \tag{42}$$

$$\begin{aligned}\frac{\partial d}{\partial\boldsymbol{\omega}} = \boldsymbol{M} &= \begin{bmatrix} 0 & \hat{p}_x\hat{\omega}_z & \hat{p}_x\hat{\omega}_y \\ \hat{p}_y\hat{\omega}_z & 0 & \hat{p}_y\hat{\omega}_x \\ \hat{p}_z\hat{\omega}_y & \hat{p}_z\hat{\omega}_x & 0 \end{bmatrix}, \\ \frac{\partial d}{\partial\boldsymbol{p}} = \boldsymbol{N} &= \begin{bmatrix} \hat{\omega}_y\hat{\omega}_z & 0 & 0 \\ 0 & \hat{\omega}_x\hat{\omega}_z & 0 \\ 0 & 0 & \hat{\omega}_x\hat{\omega}_y \end{bmatrix}\end{aligned} \tag{43}$$

The translation part of the system matrix is expressed as

$$\boldsymbol{F}_2 = \begin{bmatrix} \boldsymbol{0}_3 & \boldsymbol{I}_3 \\ -2\boldsymbol{S}(\Omega) & \boldsymbol{K}(n) \end{bmatrix} \tag{44}$$

Thus, the system matrix $\boldsymbol{F}$ is written as follows:

$$\boldsymbol{F} = \begin{bmatrix} \boldsymbol{F}_1 & \boldsymbol{0}_{10\times6} \\ \boldsymbol{0}_{6\times10} & \boldsymbol{F}_2 \end{bmatrix} \tag{45}$$

The discrete state-transition matrix is obtained using the first-order approximation as follows:

$$\boldsymbol{\Phi} \approx \boldsymbol{I}_{16} + \boldsymbol{F}\cdot\Delta T = \begin{bmatrix} \boldsymbol{\Phi}_{11} & \boldsymbol{0}_{10\times6} \\ \boldsymbol{0}_{6\times10} & \boldsymbol{\Phi}_{22} \end{bmatrix} \tag{46}$$

where ΔT is the sampling time interval.

Thus, the linearized discrete-time system dynamic can be expressed as

$$\delta\boldsymbol{x}_{e,k+1} = \boldsymbol{\Phi}\cdot\delta\boldsymbol{x}_{e,k} + \boldsymbol{\omega}_k \tag{47}$$

where $\boldsymbol{\omega}_k$ is zero-mean Gaussian process noise with covariance matrix $\boldsymbol{Q}$.

The observation equation can be written as,

$$\begin{aligned}\boldsymbol{z}_{e,k} &= h(\boldsymbol{x}_{e,k}) + \boldsymbol{v}_{e,k} \\ &= \begin{bmatrix} \boldsymbol{q}_{m,k} \\ \boldsymbol{r}_{m,k} \end{bmatrix} = \begin{bmatrix} \boldsymbol{q}_k \\ \boldsymbol{r}_k \end{bmatrix} + \boldsymbol{v}_{e,k} = \boldsymbol{H}\cdot\boldsymbol{x}_{e,k} + \boldsymbol{v}_{e,k}\end{aligned} \tag{48}$$

where, $\boldsymbol{z}_{e,k} = [\boldsymbol{q}_m^T\boldsymbol{r}_m^T]^T$ is the measurement output, $\boldsymbol{v}_{e,k}$ is the measurement noise(assumed to be zero-mean Gaussian process noise with covariance matrix $\boldsymbol{R}$), and $\boldsymbol{H}$ is the observation sensitivity matrix, as follows:

$$\boldsymbol{H} = \begin{bmatrix} \boldsymbol{I}_4 & \boldsymbol{0}_{4\times3} & \boldsymbol{0}_{4\times3} & \boldsymbol{0}_{4\times3} & \boldsymbol{0}_{4\times3} \\ \boldsymbol{0}_{3\times4} & \boldsymbol{0}_3 & \boldsymbol{0}_3 & \boldsymbol{I}_3 & \boldsymbol{0}_3 \end{bmatrix} \tag{49}$$

The state $\boldsymbol{x}_e$ is then estimated using an EKF-based observer, as follows:

a) Initialization

$$\hat{\boldsymbol{x}}_{e0}^+ = \hat{\boldsymbol{x}}_e(t_0), \boldsymbol{P}_0^+ = \boldsymbol{P}(t_0) \tag{50}$$

b) Prediction

$$\begin{bmatrix} \hat{\boldsymbol{r}}_k^- \\ \hat{\boldsymbol{v}}_k^- \end{bmatrix} = \boldsymbol{\Phi}_{22}\begin{bmatrix} \hat{\boldsymbol{r}}_{k-1}^+ \\ \hat{\boldsymbol{v}}_{k-1}^+ \end{bmatrix} + \boldsymbol{B}\cdot\Delta T\cdot\boldsymbol{u}_{k-1} \tag{51}$$

$$\hat{\boldsymbol{q}}_k^- = \hat{\boldsymbol{q}}_{k-1}^+ + \frac{1}{2}\boldsymbol{\omega}_{k-1}^+\otimes\hat{\boldsymbol{q}}_{k-1}^+\Delta T \tag{52}$$

$$\hat{\boldsymbol{q}}_k^+ = \frac{\hat{\boldsymbol{q}}_k^+}{\|\hat{\boldsymbol{q}}_k^+\|} \tag{53}$$

$$\hat{\boldsymbol{\omega}}_k^- = \hat{\boldsymbol{\omega}}_{k-1}^+ + \boldsymbol{\psi}(\hat{\boldsymbol{\omega}}_{k-1}^+,\hat{\boldsymbol{p}}_{k-1}^+)\Delta T \tag{54}$$

$$\boldsymbol{P}_k^- = \boldsymbol{\Phi}_{k-1}\boldsymbol{P}_{k-1}^+\boldsymbol{\Phi}_{k-1}^T + \boldsymbol{Q} \tag{55}$$

$$\boldsymbol{K}_k = \boldsymbol{P}_k^-\boldsymbol{H}^T(\boldsymbol{H}_k\boldsymbol{P}_k^-\boldsymbol{H}^T + \boldsymbol{R})^{-1} \tag{56}$$

c) Update

$$\delta\boldsymbol{x}_k^+ = \boldsymbol{K}_k\left[(\boldsymbol{q}_{m,k} - \boldsymbol{q}_k^-)^T(\boldsymbol{r}_{m,k} - \hat{\boldsymbol{r}}_k^-)^T\right]^T \tag{57}$$

$$\hat{\boldsymbol{q}}_k^+ = \hat{\boldsymbol{q}}_k^- + \delta\boldsymbol{q}_k, \tag{58}$$

$$\hat{\boldsymbol{q}}_k^+ = \frac{\hat{\boldsymbol{q}}_k^+}{\|\hat{\boldsymbol{q}}_k^+\|} \tag{59}$$

$$\hat{\boldsymbol{\omega}}_k^+ = \hat{\boldsymbol{\omega}}_k^- + \delta\boldsymbol{\omega}_k \tag{60}$$

$$\hat{\boldsymbol{p}}_k = \hat{\boldsymbol{p}}_{k-1} + \delta\boldsymbol{p}_k \tag{61}$$

$$\boldsymbol{r}_k^+ = \boldsymbol{r}_k^- + \delta \boldsymbol{r}_k \tag{62}$$

$$\boldsymbol{v}_k^+ = \boldsymbol{v}_k^- + \delta \boldsymbol{v}_k \tag{63}$$

Then, the state $\boldsymbol{x}_e = [\boldsymbol{q}, \boldsymbol{\omega}, \boldsymbol{p}, \boldsymbol{r}, \boldsymbol{v}]^T$ is estimated at the instant k. According to the kinematic equation of the tumbling target(Eq.(7)), the future attitude motion of the tumbling target can be predicted, and so can the trajectory of the tumbling target's docking port.

3.2. Tube-based MPC for tracking design

In this study, the control objective is to approach the neighborhood of the time-varying docking point and follow it while satisfying the engineering constraints and considering uncertainties. In addition, the trajectory of the docking port of the tumbling target is assumed to be periodic in the ECI frame. It can be predicted when the inertial parameters are estimated. The constraints of the nominal system can be obtained by tightening those of the real system using the reachable set method. Due to the tumbling characteristics of the docking port, the optimal trajectory of the chaser relative to the target may not be reachable, considering the engineering constraints of the chaser and the uncertainties of the AR&D mission. Therefore, in this study, we designed a controller that guarantees that the closed-loop system converges asymptotically and robustly to the optimal admissible periodic trajectory in an uncertain environment.

Robust trajectory planning and tracking control are generally classified into two parts. The optimal reachable trajectory is usually obtained from the trajectory planner, where a specific criterion is optimized while guaranteeing that constraint satisfaction is robustly fulfilled and then provided to the lower-layer tracking controller as a target trajectory (Limon et al., 2016). For a given period N_T, the optimal periodic reachable trajectory is determined by its initial state $\bar{\boldsymbol{x}}_0^r$ and the corresponding input sequence $\bar{\boldsymbol{U}}^r = \left(\bar{\boldsymbol{u}}^{rT}(0), \bar{\boldsymbol{u}}^{rT}(1), \ldots, \bar{\boldsymbol{u}}^{rT}(N_T-1)\right)^T$, and is the solution to the following optimization problem:

$$\min_{\bar{\boldsymbol{x}}_0^r, \bar{\boldsymbol{U}}^r} V_p(\boldsymbol{r}_t; \bar{\boldsymbol{x}}_0^r, \bar{\boldsymbol{U}}^r) = \min_{\bar{\boldsymbol{x}}_0^r, \bar{\boldsymbol{U}}^r} \left(\sum_{j=0}^{N_T-1} \| \bar{\boldsymbol{y}}^r(j) - \boldsymbol{r}(j) \|_{\boldsymbol{S}_y}^2 + \sum_{j=0}^{N_T-2} \| \bar{\boldsymbol{u}}^r(j+1) - \bar{\boldsymbol{u}}^r(j) \|_{\boldsymbol{S}_u}^2 \right) \tag{64a}$$

$$\text{s.t. } \bar{\boldsymbol{x}}^r(j+1) = \boldsymbol{A}_d \bar{\boldsymbol{x}}^r(j) + \boldsymbol{B}_d \bar{\boldsymbol{u}}^r(j) \; j = 0, \ldots, N_T \tag{64b}$$

$$\bar{\boldsymbol{x}}^r(0) = \boldsymbol{x}_0^r \tag{64c}$$

$$\bar{\boldsymbol{y}}^r(j) = \boldsymbol{C} \bar{x}^r(j) \; j = 0, \ldots, N_T \tag{64d}$$

$$(\bar{\boldsymbol{x}}^r(j), \bar{\boldsymbol{u}}^r(j)) \in \bar{\mathbb{Z}}(j) \; j = 0, \ldots, N_T \tag{64e}$$

$$\bar{\boldsymbol{y}}^r(j) \in \Gamma_{kz}^{(j)} j = 0, \ldots, N_T \tag{64f}$$

$$\bar{\boldsymbol{x}}^r(T) = \bar{\boldsymbol{x}}^r(0) \tag{64g}$$

where the term $(\bar{\boldsymbol{x}}^r(j), \bar{\boldsymbol{u}}^r(j)) \in \bar{\mathbb{Z}}(j)$ denotes the general engineering constraint. The cost function penalizes the deviation between the reachable trajectory and the target reference and the increment of the planned admissible input. Though minimizing the cost function, the reachable/planned trajectory converges to the reference trajectory while making the former trajectory smoother and more accessible for onboard implementation. The matrices $\boldsymbol{S}_y$ and $\boldsymbol{S}_u$ are symmetric and positively defined. The optimal or near-optimal admissible trajectory $\boldsymbol{y}^{r*}(j)$ is obtained by solving the optimization problem (Eq.(64)).

However, this classical method may not be efficient for onboard implementation when the target/reference is time-varying and computation capacity is limited, which is the case with the tumbling target considered in this study. Therefore, a novel control formulation is proposed, combining trajectory planning and tracking control in a single optimization layer based on MPC for tracking theory, which is compatible with a feedback control strategy that robustly ensures constraint satisfaction in an uncertain environment. In contrast to previous tracking MPC serving as a low-layer controller to track the planned trajectory, the proposed novel control formulation unites trajectory planning and control in a signal layer. In the novel control formulation, the cost function $V_{N_p}(\boldsymbol{x}_k, \boldsymbol{r}_k; \bar{\boldsymbol{x}}^r, \bar{\boldsymbol{U}}^r, \bar{\boldsymbol{U}}(k))$ penalizes the following terms: (i)$V_t(\boldsymbol{x}_k; \bar{\boldsymbol{x}}^r, \bar{\boldsymbol{U}}^r, \bar{\boldsymbol{U}}(k))$, the deviation in the evolution of the nominal system (Eq. (24)) concerning the planned reachable reference trajectory along the prediction horizon N_p; and (ii)$V_p(\boldsymbol{r}_k; \bar{\boldsymbol{x}}^r, \bar{\boldsymbol{U}}^r)$, the distance from the planned reachable trajectory to the reference trajectory of the docking port for one period N_T. Therefore, the cost function $V_{N_p}(\boldsymbol{x}_k, \boldsymbol{r}(k); \bar{\boldsymbol{x}}^r, \bar{\boldsymbol{u}}^r, \bar{\boldsymbol{U}}(k))$ is given by

$$V_{N_p}(\boldsymbol{x}_k, \boldsymbol{r}_k; \bar{\boldsymbol{x}}^r, \bar{\boldsymbol{U}}^r, \bar{\boldsymbol{U}}(k)) = V_t(\boldsymbol{x}_k; \bar{\boldsymbol{x}}^r, \bar{\boldsymbol{U}}^r, \bar{\boldsymbol{U}}(k)) + V_p(\boldsymbol{r}_k; \bar{\boldsymbol{x}}^r, \bar{\boldsymbol{U}}^r) \tag{65}$$

The parameters $(\boldsymbol{x}_k, \boldsymbol{r}_k)$ represent the chaser's current state and reference trajectory to be tracked, respectively. The decision variables are a planned periodic reachable trajectory defined by its initial state $\bar{\boldsymbol{x}}^r$ and the corresponding sequence of control inputs $\bar{\boldsymbol{U}}^r$, as well as a sequence of nominal open-loop control inputs $\bar{\boldsymbol{U}}(k)$. According to the above analysis, the terms $V_t(\boldsymbol{x}_k; \bar{\boldsymbol{x}}^r, \bar{\boldsymbol{U}}^r, \bar{\boldsymbol{U}}(k))$ and $V_p(\boldsymbol{r}_k; \bar{\boldsymbol{x}}^r, \bar{\boldsymbol{U}}^r)$ are defined as,

$$V_t(\boldsymbol{x}_k; \bar{\boldsymbol{x}}^r, \bar{\boldsymbol{U}}^r, \bar{\boldsymbol{U}}(k)) = \sum_{i=0}^{N_p} \left(\| \bar{\boldsymbol{x}}(k+i|k) - \boldsymbol{x}^r(i) \|_{\boldsymbol{Q}}^2 + \| \bar{\boldsymbol{u}}(k+i|k) - \boldsymbol{u}^r(i) \|_{\boldsymbol{R}}^2 \right) \tag{66}$$

$$V_p(\boldsymbol{r}_k;\bar{\boldsymbol{x}}^r,\bar{\boldsymbol{U}}^r)=\sum_{i=0}^{N_T}\|\bar{\boldsymbol{y}}^r(i)-\boldsymbol{r}_k(i)\|^2_{\boldsymbol{S}_y}+\sum_{i=0}^{N_T}\|\bar{\boldsymbol{u}}^r(i+1)-\bar{\boldsymbol{u}}^r(i)\|^2_{\boldsymbol{S}_u} \quad (67)$$

where $\boldsymbol{S}_u$ and $\boldsymbol{S}_y$ are symmetric and positive-definite matrices. Penalization of the increment of the planned admissible input $\|\bar{\boldsymbol{u}}^r(i+1)-\bar{\boldsymbol{u}}^r(i)\|^2_{\boldsymbol{S}_u}$ is incorporated into the cost function to force the planned reachable trajectory to be smoother. Trajectory planning and control are then designed based on a single layer by solving the optimization problem $P_{N_p}(\boldsymbol{x}_k,\boldsymbol{r}_k)$ at each sampling instant, as follows:

$$\min_{\boldsymbol{x}^{-r},\boldsymbol{U}^{-r},\boldsymbol{U}^{-}(k)} V_{N_p}(\boldsymbol{x}_k,\boldsymbol{r}(k);\ \bar{\boldsymbol{x}}^r,\bar{\boldsymbol{U}}^r,\bar{\boldsymbol{U}}(k)) \quad (68a)$$

$$\text{s.t.}\boldsymbol{x}_k=\bar{\boldsymbol{x}}(k|k)\oplus\mathbb{W} \quad (68b)$$

$$\bar{\boldsymbol{x}}(k+i+1|k)=\boldsymbol{A}_d\bar{\boldsymbol{x}}(k+i|k)+\boldsymbol{B}_d\bar{\boldsymbol{u}}(k+i|k),\ i=0,\dots,N_p-1 \quad (68c)$$

$$\bar{\boldsymbol{y}}(k+i|k)=\boldsymbol{C}\bar{\boldsymbol{x}}(k+i|k),i=0,\dots,N_p \quad (68d)$$

$$(\bar{\boldsymbol{x}}(k+i|k),\bar{\boldsymbol{u}}(k+i|k))\in\bar{\mathbb{Z}},i=0,\dots,N_p \quad (68e)$$

$$\bar{\boldsymbol{y}}(k+i|k)\in\Gamma_{kz}^{(k+i)},i=0,\dots,N_p \quad (68f)$$

$$\bar{\boldsymbol{x}}^r(0)=\bar{\boldsymbol{x}}^r \quad (68g)$$

$$\bar{\boldsymbol{x}}^r(j+1)=\boldsymbol{A}_d\bar{\boldsymbol{x}}^r(i)+\boldsymbol{B}_d\bar{\boldsymbol{u}}^r(i),j=0,\dots,N_T-1 \quad (68h)$$

$$\bar{\boldsymbol{y}}^r(j)=\boldsymbol{C}\bar{\boldsymbol{x}}^r(j),i=0,\dots,N_T \quad (68i)$$

$$(\bar{\boldsymbol{x}}^r(j),\bar{\boldsymbol{u}}^r(j))\in\bar{\mathbb{Z}},i=0,\dots,N_T \quad (68j)$$

$$\bar{\boldsymbol{y}}^r(j)\in\Gamma_{kz}^{(j)},i=0,\dots,N_T \quad (68k)$$

$$\bar{\boldsymbol{x}}(k+N_p|k)=\bar{\boldsymbol{x}}^r(N_p) \quad (68l)$$

Let the symbol $V_{N_p}(\bar{x}^{r*},\bar{\boldsymbol{u}}^{r*},\bar{\boldsymbol{U}}^*(k))$ denote the optimal solution to this optimization problem (Eq.(68)). The optimal predicted nominal state trajectory is denoted as $\bar{\boldsymbol{X}}^*(\boldsymbol{x_k},\boldsymbol{r}_k)$ when the optimal control input sequence $\bar{\boldsymbol{U}}^*(k)$ is implemented on the nominal system, while the symbol $\bar{\boldsymbol{y}}^*_{N_p}(\boldsymbol{x}_k,\boldsymbol{r}_k)$ denotes the corresponding output trajectory of the nominal system. The optimal planned periodic admissible/reachable trajectory is denoted as $\bar{\boldsymbol{x}}^{r*}(\boldsymbol{x}_k,\boldsymbol{r}_k)$, while $\bar{\boldsymbol{y}}^{r*}(\boldsymbol{x}_k,\boldsymbol{r}_k)$ represents the corresponding output trajectory. The stability and feasibility of the controller are guaranteed by the design of the terminal condition(Eq.(68l)). The optimization problem (Eq.(68)) is a convex optimization that many methods, such as the interior point and active set methods, can solve efficiently.

Proposition 1. Suppose $\mathbb{H}(i)$ is the reachable set for a real system $\boldsymbol{x}(i+1)=\boldsymbol{A}_k\boldsymbol{x}(i)+\boldsymbol{\omega}$, for $i=0,\dots,N_p$, where $\boldsymbol{A}_k=\boldsymbol{A}_d+\boldsymbol{B}_d\boldsymbol{K}$ and $\boldsymbol{K}$ is chosen such that $\boldsymbol{A}_k$ is stable. If $\boldsymbol{x}(i)\in\bar{\boldsymbol{x}}(i)\oplus\mathbb{H}(i)$, and $\boldsymbol{u}(i)=\bar{\boldsymbol{u}}(i)+\boldsymbol{K}(\boldsymbol{x}(i)-\bar{\boldsymbol{x}}(i))$, then $\boldsymbol{x}(i+1)\in\bar{\boldsymbol{x}}(i+1)\oplus\mathbb{H}(i+1)$ holds for all $\boldsymbol{\omega}\in\mathbb{W}$, where $\boldsymbol{x}(i+1)=\boldsymbol{A}_d\boldsymbol{x}(i)+\boldsymbol{B}_d\boldsymbol{u}(i)+\boldsymbol{\omega},\bar{x}(i+1)=\boldsymbol{A}_d\bar{x}(i)+\boldsymbol{B}_d\bar{u}(i)$.

Inspired by **Proposition 1**, the control law of the proposed controller consists of the feedforward control $\bar{u}^*(k|k)$ and the feedback control, as follows:

$$\kappa_{N_p}(k)=\bar{\boldsymbol{u}}^*(k|k)+\boldsymbol{K}(\boldsymbol{x}_k-\bar{\boldsymbol{x}}^*(k|k)) \quad (69)$$

where $\bar{\boldsymbol{u}}^*(k|k)$ is the first control input of the optimal control sequence $\bar{\boldsymbol{U}}^*(k)$.

Remark 1. The proposed robust controller uses a reachable set to quantify the propagation of uncertainties instead of the minimum robust positive invariant (mRPI) set; this approach could enlarge the solution space of the decision variables and reduce the conservatism of the robust control algorithm.

Remark 2. The proposed robust controller is designed based on a single layer that unites dynamic trajectory planning and control, fully utilizing the periodic information of the target's position, and the optimization is a simple QP problem of similar complexity to classical MPC.

Remark 3. The controller is stable by design and can drive the output of the nominal system to converge asymptotically to the optimal admissible periodic trajectory. It can also force every realization of the actual states in a reachable set centered on the nominal one. In addition, the constraints of the optimization problem solved by the controller do not depend on the reference, allowing for a sudden change in the reference without losing feasibility. Thus, the controller still functions when the reference is quasi-periodic.

4. Simulation results

4.1. Scenario configuration and parameter settings

We assume that the chaser's sensor initially points to the target. The target is rotating around one of its principal axes. Meanwhile, the principal axis revolves around the direction of the angular momentum $\boldsymbol{H}$. Table 1 lists the detailed simulation parameters of the target. The relative position of the chaser spacecraft (x_0,y_0,z_0) is initially $(-8,15,16)$ m, as in the reference (Li et al., 2017). The relative velocity of the chaser is set as $(x_0',y_0',z_0')=(0,0,0)$ m/s, since the AR&D problem is a close-range problem. In this study, the thruster directions align with the body axes. Each direction has four pairs of thrusters, each providing a force of ± 5 N with a chaser mass of 100 kg. Therefore, the maximum control $\boldsymbol{u}_{\max}$ is limited to a maximum magnitude of 0.2 m/s^2 (Flores-Abad et al., 2014). Each direction of the maximum approach velocity $\boldsymbol{v}_{\max}$ is constrained to 0.5 m/s to ensure the safety of the AR&D mission in the case of an emergency. The radius

Table 1
Simulation parameters of the target.

Parameter	Value	Unit
Semi-major axis,a	7,178,160	m
Eccentricity,e	0.003	
Inclination,i	60	deg
Longitude of the ascending node,Ω	30	deg
Argument of latitude,ϑ	0	deg
Moment of inertia,I_T	$\mathrm{diag}(100, 100, 60)$	$\mathrm{kg \cdot m^2}$
Docking port,r_{doc}	(0,0,1)	m
Initial angles,(ψ_0, φ_0)	(0,0)	deg
Initial rotation rate,$(\omega_{x_0}^{BF}, \omega_{y_0}^{BF}, \omega_{z_0}^{BF})$	(0.27,1.96,1.90)	$\mathrm{deg \cdot s^{-1}}$
Magnitude of angular momentum,$\| \mathbf{H} \|_2$	3.14	$\mathrm{kg \cdot m^2 \cdot s^{-1}}$
Critical distance,r_s	1	m
Nutation angle,θ	60	deg

of the "keep-out" zone that avoids collision is set as $r_s = 1$ m. The external disturbance $\boldsymbol{d}$ that determines the bounded set $\mathbb{W}_1$ is set as

$$\boldsymbol{d} = \begin{pmatrix} \sin(\omega_l t) \\ 2\sin(\omega_l t + 0.5\pi) \\ 1.5\cos(\omega_l t) \end{pmatrix} \times 10^{-3} \tag{70}$$

where the constant is set as $\omega_l = 0.05$.

The bound for the control deviation, which is related to the uncertainty set $\mathbb{W}_2$, is set as 5% of the maximum control magnitude, i.e., $\bar{\alpha} = 5\%$. The accuracy of the measurement sensors in terms of position and velocity is 1% of its measurement range, easily yielding the uncertainty set $\mathbb{W}_3$. Then, the total uncertainty set $\mathbb{W}$ can be calculated using Eq. (22).

To test the efficiency of the proposed control method, the real system's states are updated from the model of the nonlinear dynamics (Eq. (2), Dong et al., 2020) via fourth-order Runge-Kutta numerical integration, while the proposed controller is designed based on the C-W equations (Eq.(1)). A computer using an Intel® CPU @ 2.50 GHz with 8 GB RAM and MATLAB 2016 was used as the simulation environment, along with the YALMIP suite and the MOSEK toolbox as the solver. The expected tracking error regarding each direction's relative position and velocity in the LVLH frame is less than 0.2 m and 0.1 m/s, respectively, and the maximum time-to-dock is 60 s. The computation time was measured by CPU time consumption.

To thoroughly verify the effectiveness of the proposed control method and consider the difficulty level of the AR&D mission, two scenarios were investigated. In scenario 1, we assumed that the target was not tumbling, and in scenario 2, the target was tumbling.

4.2. Scenario 1: The target is not tumbling

This study first analyzes the simple scenario in which the target is not tumbling; that is, the relative position of the docking axis of the target remains unchanged in the Hill coordinate system. Fig. 4 describes the 3-D relative motion trajectory of the chaser. Figs. 5 and 6 illustrate the time histories of the chaser's relative position and distance, respectively. Fig. 7 shows the time history of the chaser's control acceleration.

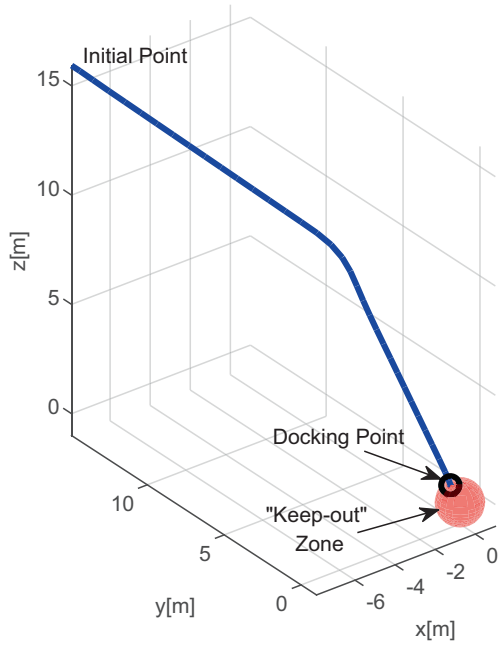


Fig. 4. 3-D trajectory of the chaser when the target is not tumbling.

In Fig. 4, the blue dotted line represents the relative motion trajectory of the chaser, and the light red area is the "keep-out" zone. Fig. 4 shows that the chaser successfully reached the vicinity of the docking point of the target from its initial position, and its 3-D trajectory was smooth considering the uncertainty, thus allowing for easy engineering implementation.

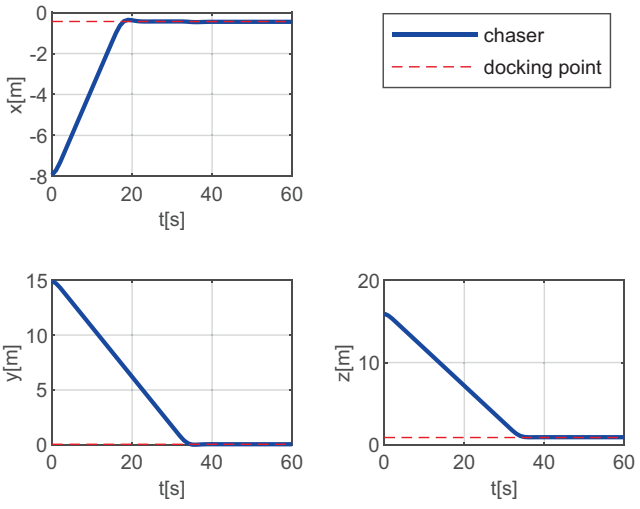


Fig. 5. Time history of the relative position of the chaser when the target is not tumbling.

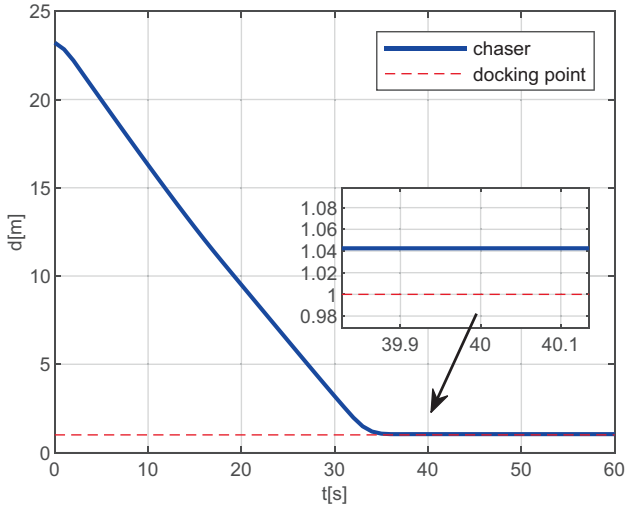


Fig. 6. Time history of the relative distance of the chaser when the target is not tumbling.

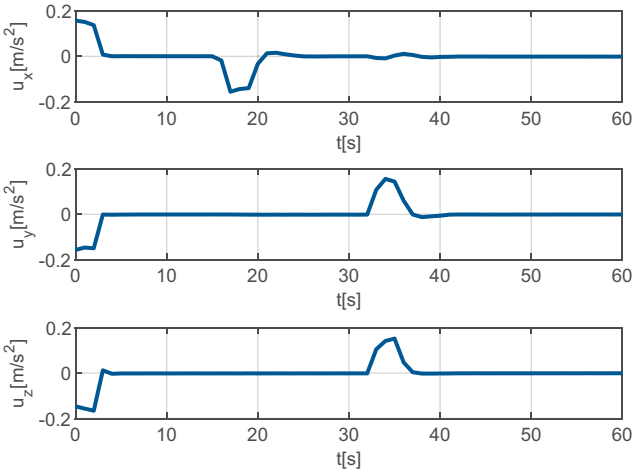


Fig. 7. Time history of the control input of the chaser when the target is not tumbling.

In Figs. 5–7, the solid blue lines represent the trajectory of the position and distance of the chaser relative to the center of the target, and the red dashed lines represent the trajectory of the relative position and distance of the target's docking point. Fig. 5 shows that the controller drives the chaser to approach the chaser's docking point, and it finally reaches the vicinity of the docking point at approximately 35 s. Fig. 6 shows that the distance between the chaser and the target's docking point gradually decreased and finally converged to a small range (within 0.05 m). In addition, the enlarged drawing of Fig. 6 shows that under uncertainty, the chaser approaches the docking axis from the outside direction of the "keep-out" zone, meeting the collision avoidance constraint to guarantee the safety of the AR&D mission. Fig. 7 shows that the designed controller's control acceleration curve is smooth and easy to implement in practice and satisfies the control saturation constraint (within 0.2 m/s^2).

In summary, the TMPT controller designed in this study can successfully perform the AR&D mission while satisfying the engineering constraints in the presence of multiple uncertainties when the space target is not tumbling. Compared with a traditional robust MPC, the stability and the feasibility of the proposed TMPT controller of the integral design architecture can be achieved with a smaller prediction horizon up to the system order. Thus, it could lower the computation complexity of the control algorithm and make it suitable for online implementation. Moreover, the constraints in the proposed TMPT controller are independent of the reference signal, allowing for sudden changes in the reference without losing feasibility and stability.

4.3. Scenario 2: The target is tumbling

In this scenario, we assume that the target is tumbling freely in space. To thoroughly verify the effectiveness and necessity of the proposed unified framework, which uses an integrated design of identification, prediction, planning, and control for the AR&D mission with uncertainties, the robust MPC for tracking (TMPT) method without estimation was tested first and then compared with the proposed unified framework.

4.3.1. Simulation result of the TMPT method without estimation

When the TMPT controller without estimation is designed to approach the tumbling target, the docking point's current position could be considered a constant reference signal to be tracked with the "greedy strategy" because the future behavior of the target cannot be predicted.

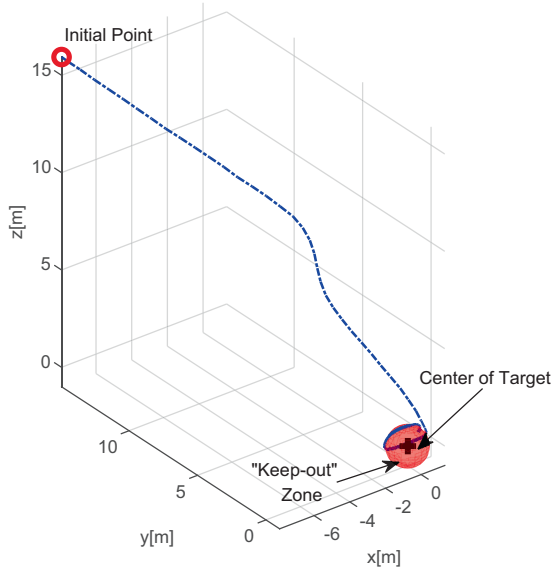


Fig. 8. 3-D trajectory of the chaser, TMPT method without estimation.

Fig. 8 describes the 3-D relative motion trajectory of the chaser using the TMPT method, Figs. 9 and 10 describe the chaser's position and distance in the target-centered LVLH frame, and Fig. 11 illustrates the chaser's control input. In Fig. 8, the blue dotted line represents the 3-D relative motion trajectory of the chaser, the symbol "+" represents the center of the tumbling target, and the light red area is the "keep-out" zone. As shown in Fig. 8, the chaser successfully reached the vicinity of the docking point of the target from its initial position and maintained synchronization with it thereafter concerning the relative position. However, in this case, the 3-D relative motion trajectory of the chaser was more tortuous compared to the situation in which the target was not tumbling (Fig. 4). In Figs. 9 and 10, the solid blue lines represent the time histories of the relative position and distance of the chaser, respectively.

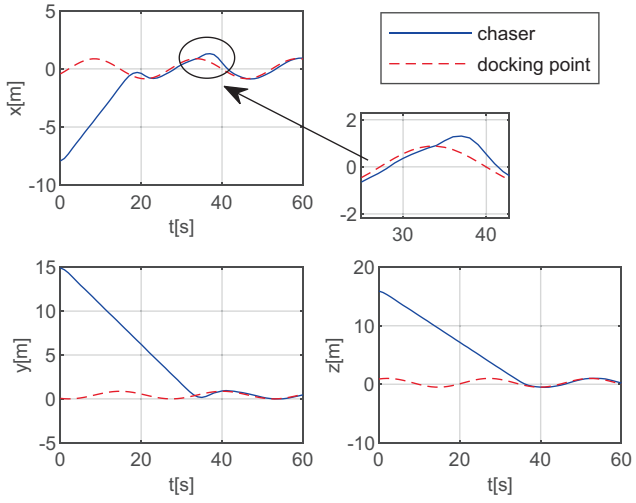


Fig. 9. Time history of the relative position of the chaser, TMPT method without estimation.

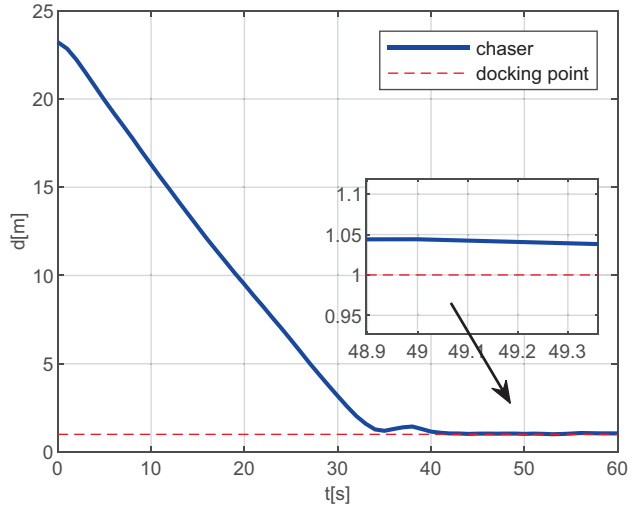


Fig. 10. Time history of the relative distance of the chaser, TMPT method without estimation.

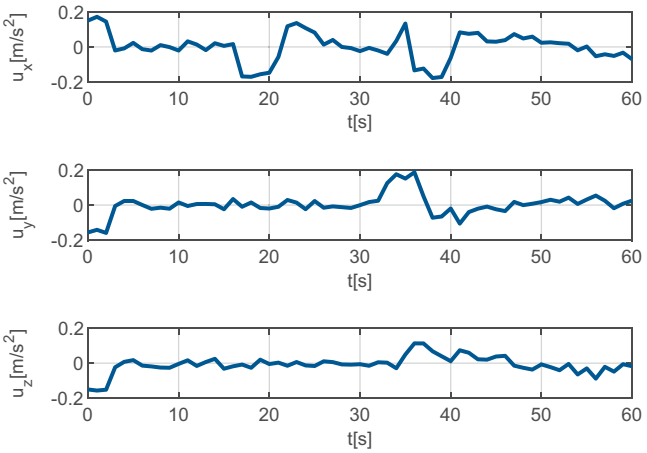


Fig. 11. Time history of the control input of the chaser, TMPT method without estimation.

The red dashed lines represent those of the docking points of the tumbling target. Figs. 9 and 10 show that the controller can drive the chaser to approach the docking point of the tumbling target and satisfy the obstacle avoidance constraint during the AR&D mission. However, it took approximately 40 s to reach the vicinity of the docking point and achieve proximity operations, representing a more time-consuming operation compared to that in scenario 1. In addition, Fig. 9 shows that a phase difference exists between the reference signal and the actual trajectory of the chaser, indicating that the tracking accuracy is low. Fig. 11 illustrates that although the control input of the designed controller meets the control saturation constraint (within 0.2 m/s^2), the curve is quite tortuous and not conducive to engineering implementation.

Although the TMPT method without estimation can drive the chaser to reach the vicinity of the target's docking point, it performs poorly in terms of time consumption and tracking accuracy while meeting engineering constraints in the presence of uncertainties. Because of its inability to predict the future behavior of the tumbling target, there exists a significant phase difference concerning the relative position between the trajectory of the chaser and the reference signal. Therefore, the TMPT method without estimation is unsuitable for AR&D with a tumbling target with uncertainties.

4.3.2. Simulation result of the proposed unified framework

To demonstrate the robustness of the proposed method in an uncertain environment with engineering constraints during the AR&D mission, we conducted 50 Monte Carlo simulations.

Fig. 12 describes the 3-D relative motion trajectory of the chaser when using the unified control framework applied in AR&D with a tumbling target. Figs. 13–15 depict the time histories of the chaser's relative position, distance, and velocity, respectively. Fig. 16 describes the control profiles of the chaser. Figs 17,19, and 21 depict

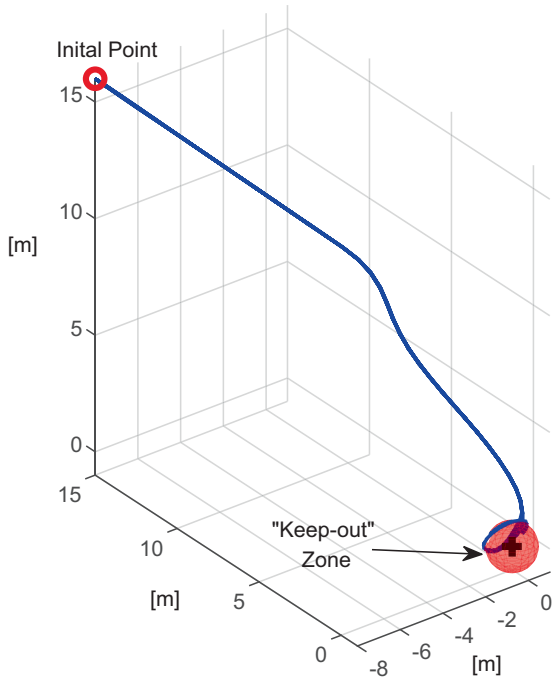


Fig. 12. 3-D trajectories of the chaser, proposed method.

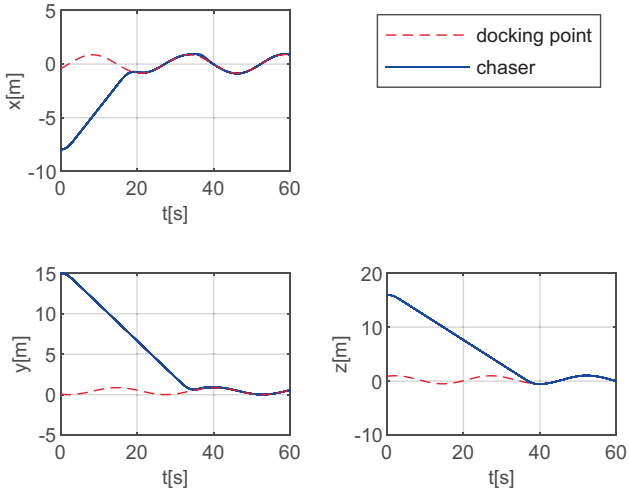


Fig. 13. Time history of the relative position of the chaser, proposed method.

the time histories of the estimation errors of the inertial parameters, attitude quaternion, and angular velocity of the tumbling target, while Figs. 18, 20, and 22 represent corresponding enlarged images in the last 20 s. Figs. 23 and 24 depict the estimation error from the chaser's actual relative position and velocity relative to the estimated values.

In Fig. 12, the blue lines represent the 3-D relative motion trajectories of the chaser after 50 times Monte Carlo simulations were performed, the symbol "+" represents the center of mass of the tumbling target, and the light red area is the "keep-out" zone. Fig. 12 shows that

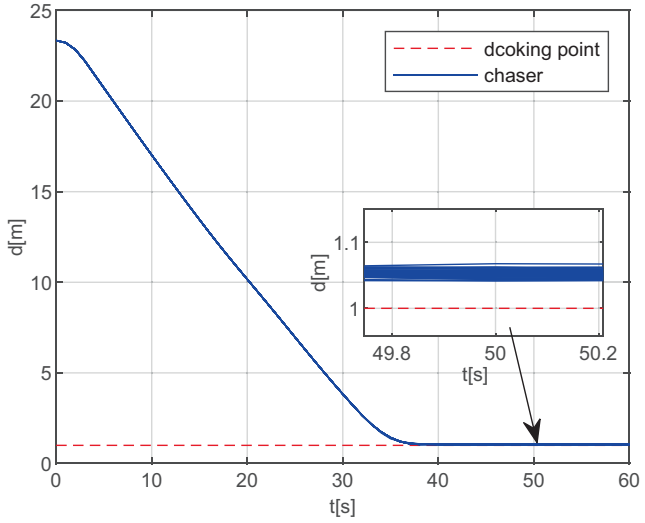


Fig. 14. Time history of the relative distance of the chaser, proposed method.

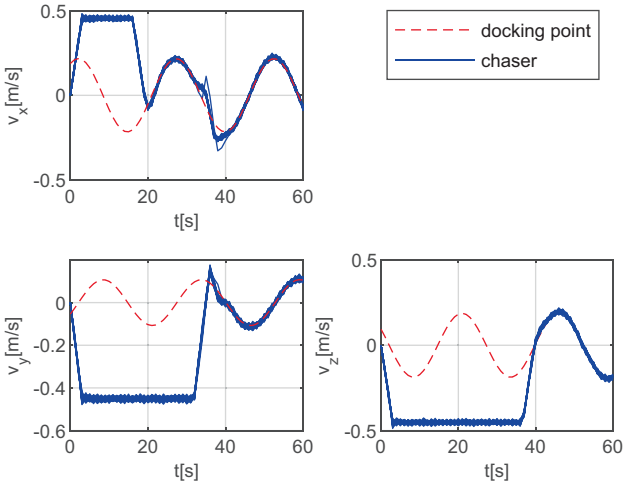


Fig. 15. Time history of the relative velocity of the chaser, proposed method.

the controller can generate a smooth and near-optimal trajectory, which can drive the chaser from the initial position to the vicinity of the target's docking point. In Figs. 13–15, the solid blue lines represent the time histories of the chaser's relative position, distance, and velocity, respectively. The red dashed lines represent the corresponding parts of the target docking point. Figs. 13–15 show that the controller drove the chaser to approach the docking point of the tumbling target. The chaser finally reached the vicinity of the docking point of the target at approximately 37 s, then maintained synchronization with the docking point in terms of relative position and velocity thereafter, demonstrating its high efficiency and superiority to the TMPT method (i.e., direct tracking control without estimation) in terms of time consumption. In addition, Fig. 13 shows that the proposed unified control framework had no phase

difference between the actual position trajectory of the chaser and the reference signal (the docking point of the target), thus outperforming the TMPT method without estimation in terms of tracking accuracy. Moreover, from the partially enlarged image in Fig. 14, we can see that the chaser approached the docking point of the chaser from the outside direction of the "keep-out" zone, showing it meets the collision avoidance constraint to ensure the safety of the AR&D mission. Fig. 16 shows that the control input satisfies the control saturation constraint (within 0.2 m/s^2) for all 50 Monte Carlo simulations. Figs. 12–16 demonstrate that the proposed unified control framework is robust to uncertainties and performs the AR&D mission successfully. An EKF was designed in this study to estimate the inertial parameters, attitude quaternion, and angular velocity. Figs. 17 and 18, show that the estimation errors of the inertial parameters ultimately converged to within 0.002 rad in the case of significant initial estimation errors, thus laying a good foundation for predicting the

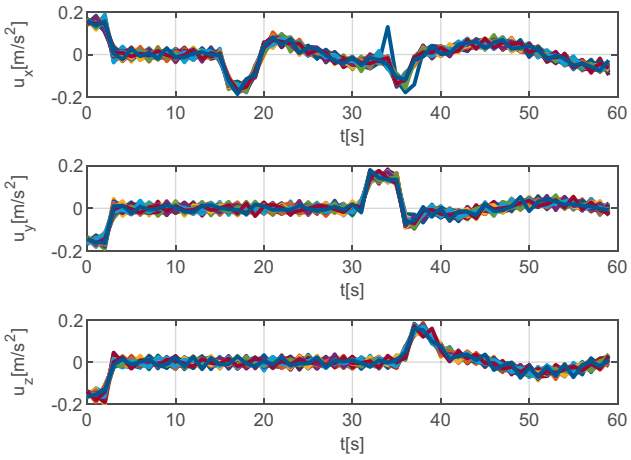


Fig. 16. Time history of the control acceleration of the chaser, proposed method.

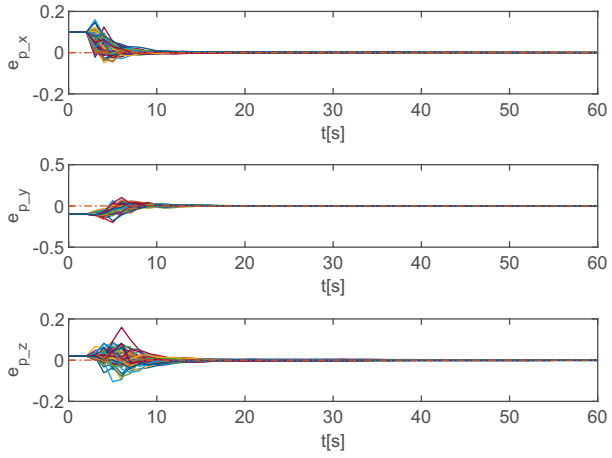


Fig. 17. Time history of the estimation error of the inertial parameters.

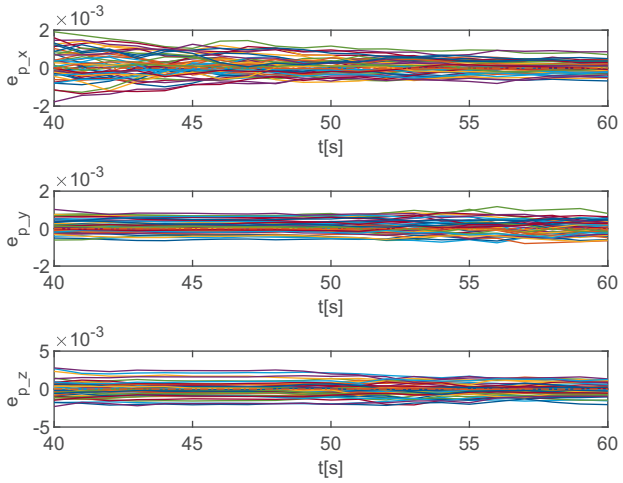


Fig. 18. Enlarged drawings of the Fig. 17 (40–60 s).

future behavior of tumbling targets, as well as the integrated design of guidance and control.

Figs. 19–22 show that the attitude quaternion and angular velocity errors ultimately converged to a small range, with the attitude quaternion estimation accuracy reaching 0.002 rad and the angular velocity accuracy reaching 0.001 rad/s. Figs. 23 and 24 show that the chaser's relative position and relative velocity tracking errors gradually converged to a small error band, with a relative position estimation accuracy of 0.01 m and a relative velocity estimation accuracy of 0.01. Figs. 17–24 show that the EKF designed in this study can effectively estimate the system's state and achieve high accuracy during the AR&D mission. The average online computation cost for solving the convex optimization problem(Eq.(68)) is 0.12 s in the aforementioned simulation environment, which is easy and ideal for online implementation.

To test the robustness of the proposed control method fully, 12 different instances are assessed under different levels of uncertainty as shown in Fig. 25. The control devi-

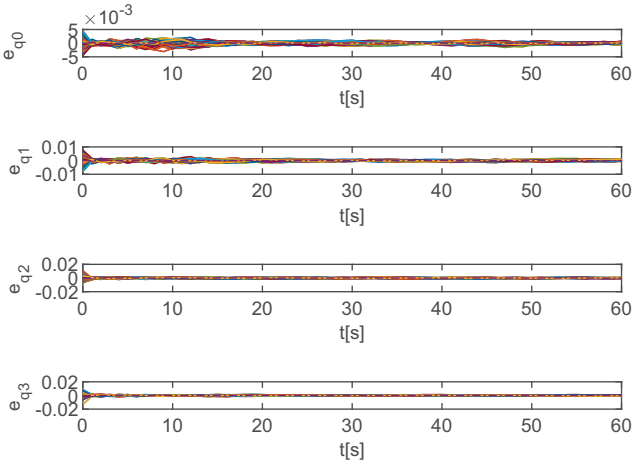


Fig. 19. Time history of the estimation error of the attitude quaternion.

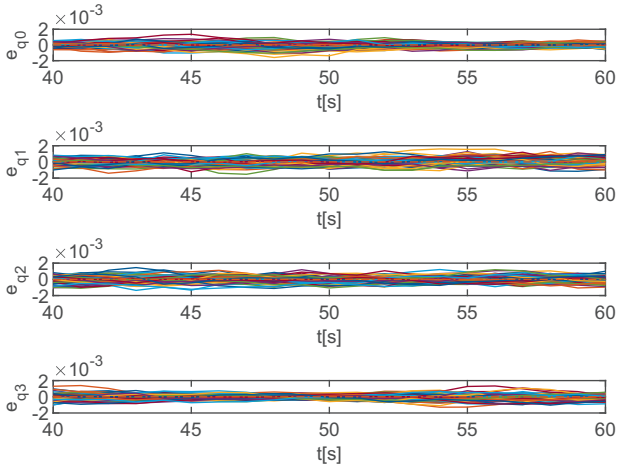


Fig. 20. Enlarged drawings of the Fig. 19 (40–60 s).

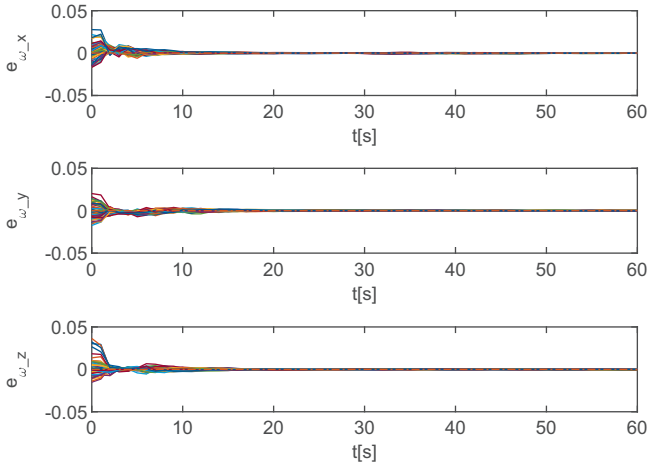


Fig. 21. Time history of the estimation error of the angular velocity.

ation ratio α is set between 0.05 and 0.2. The measurement error ratio β is set between 0 and 0.04. The criterion for assessing the success of an instance is that the mean position tracking error is less than 0.2 m in the last 10 s. The simulation results are shown in Fig. 25, and it illustrates whether the A&RD missions succeeded or not. A one represents a successful mission, and a zero represents a failed mission. Fig. 25 shows that the proposed unified control can achieve a safe online approach where the velocity, control ability, and obstacle avoidance constraints are satisfied under the uncertainty pair $\{\alpha = 0.1, \beta = 0.02\}$, which the current measurement equipment and thruster can provide.

In summary, the integrated design method of identification, prediction, trajectory planning, and control proposed in this paper can complete AR&D with a tumbling target. The simulation results show that the proposed method outperforms the TMPT method in terms of tracking accuracy and time consumption, considering the aforementioned multiple uncertainties and engineering constraints. Simulation results show that the proposed integral design architecture is robust to uncertainties and can perform the

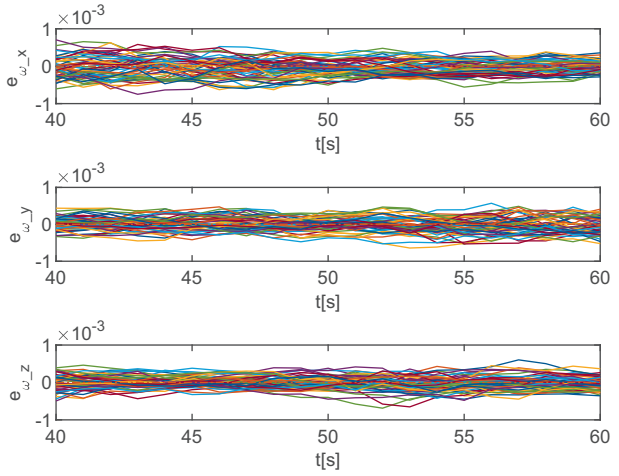


Fig. 22. Enlarged drawings of the Fig. 21 (40–60 s).

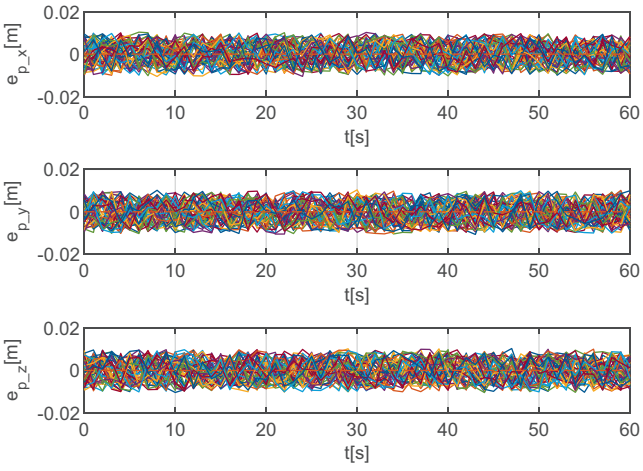


Fig. 23. Time history of the estimation error of the relative position.

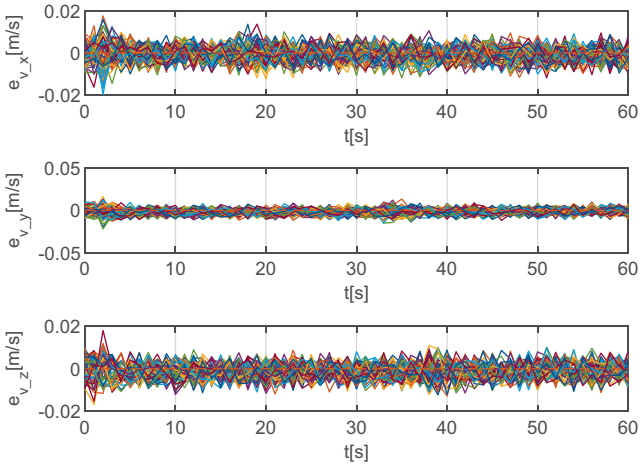


Fig. 24. Time history of the estimation error of the relative velocity.

AR&D with a tumbling target successfully. Moreover, it outperforms the TMPT method without estimation related to tracking accuracy and time consumption, considering

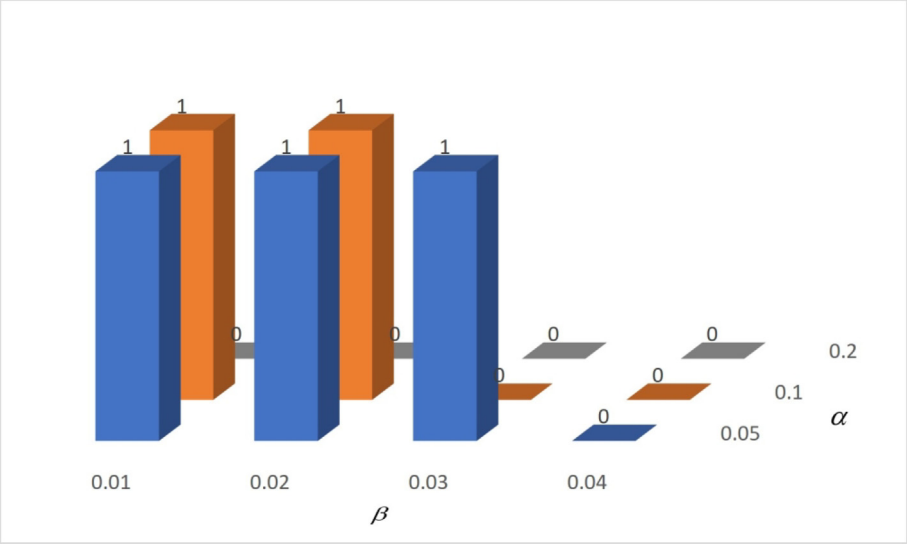


Fig. 25. Tracking performance of the proposed method subject to different levels of uncertainties.

multiple uncertainties and engineering constraints, since the former can provide the predictive information (i.e., reference signal) for the controller.

5. Conclusions

This paper proposes a unified framework for identification, prediction, motion planning, and control designed for AR&D with a tumbling target, considering engineering constraints and multiple uncertainties. The unified framework incorporates an EKF with a TMPT controller. The EKF method can effectively estimate the system states, enabling accurate prediction of the future behavior of the tumbling target based on its estimated state. The TMPT controller is robust to uncertainties and designed with a shorter prediction horizon (the same as system order) while the stability and feasibility are guaranteed, which has low computational complexity and is suitable for online and real-time implementation.

The unified control framework proposed in this paper is an integrated design of identification, prediction, motion planning, and control, representing a robustness-to-optimality control strategy. Simulations show that the unified control framework can generate a near-optimal trajectory, effectively reduce uncertainties during the control process, and accomplish the AR&D mission with high efficiency; it can eliminate phase differences between the actual position trajectory of the chaser and the reference signal; it can achieve a safe on-line approach where the velocity, control ability, and obstacle avoidance constraints are satisfied under 10 % control deviation and 2 % measurement uncertainty, which the current measurement equipment and thruster can provide. In the end, the unified control framework guarantees the closed-loop control system is stable and recursively feasible, and converges asymptotically and robustly to the optimal admissible periodic trajectory.

CRediT authorship contribution statement

Kaikai Dong: Writing – original draft, Software, Investigation. **Jianjun Luo:** Supervision, Project administration, Conceptualization. **Zhiyu Ni:** Writing – review & editing, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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